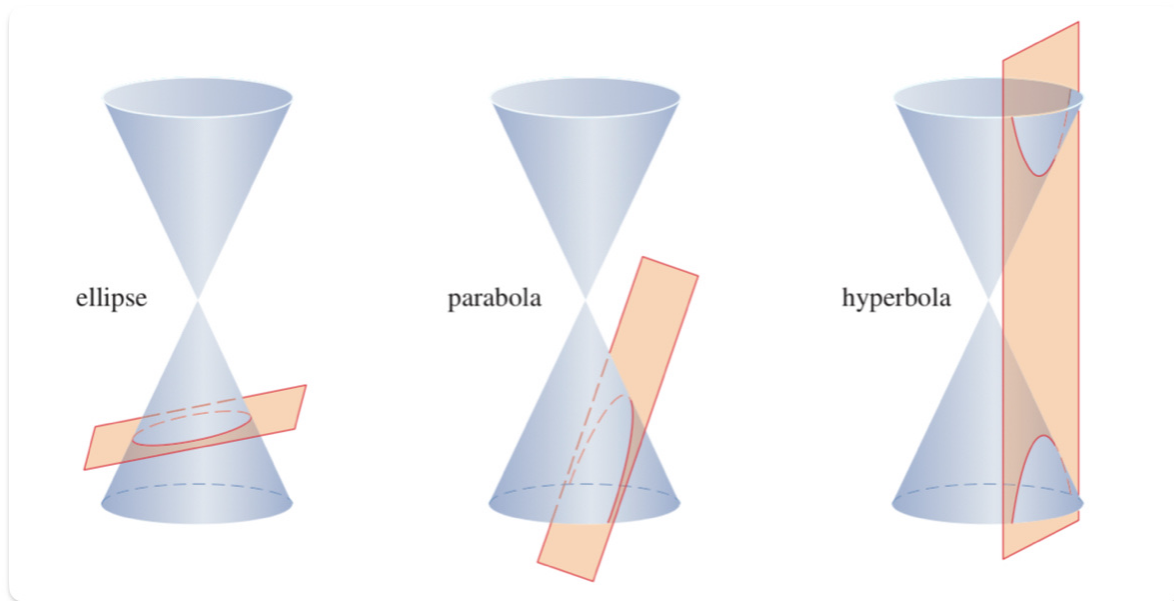


Parabola, ellipses and hyperbola are called conic sections because they result from intersecting a cone with a plane:

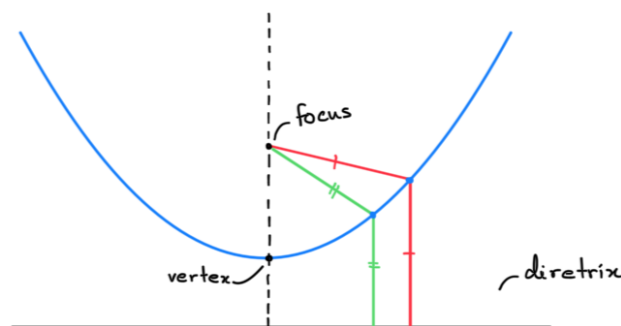


Parabola: Any point on a parabola is equidistant from a certain point and a certain line.

↳ The point is called the focus of the parabola.

↳ The line is called the diretrix of the parabola.

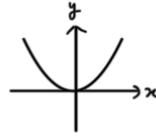
↳ The vertex is a point on the parabola which intersects the line which is perpendicular to the diretrix and passes through the focus.



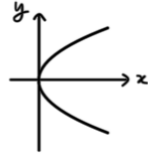
← both red lines are of the same length and both green lines are also of the same length

* Curves shaped like these are called parabolae.

So $y=x^2$ is also a parabola:

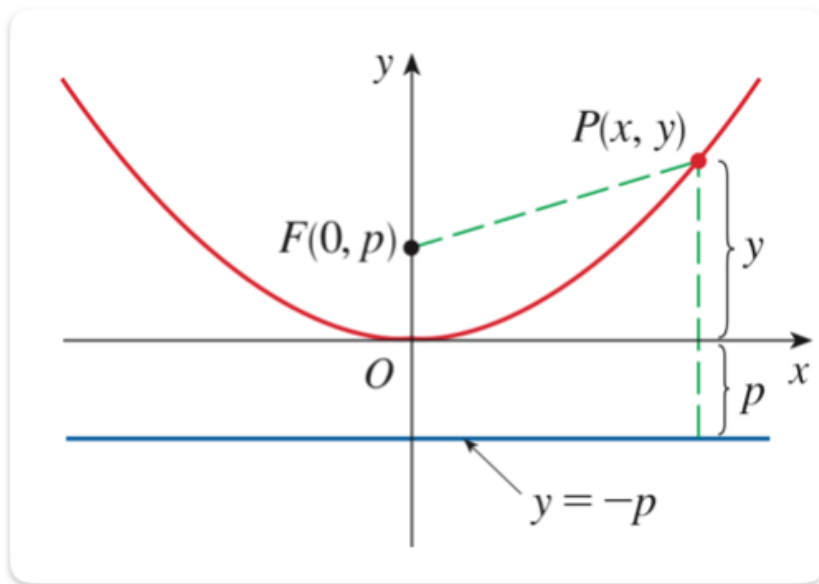


and so is the curve $x=y^2$:



* The dotted vertical line is also sometimes called the axis of the parabola.

→ Consider a simple parabola which passes through the point (0,0):



← The red curve here is the parabola and the blue line is the directrix.

↳ To derive the equation of the parabola:

Let $P(x, y)$ be a point on the parabola,

$F(0, p)$ be the focus of the parabola

and $y = -p$ be the directrix. Then:

Distance between P and F = Distance between P and $y = -p$.

$$\Rightarrow \sqrt{(x-0)^2 + (y-p)^2} = (\text{Distance between } P \text{ and the } x\text{-axis})$$

$$+ (\text{Distance between the } x\text{-axis and } y=-p)$$

$$\Rightarrow \sqrt{x^2 + y^2 - 2py + p^2} = y + p$$

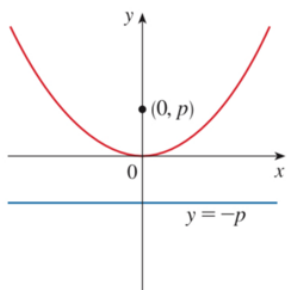
$$\Rightarrow x^2 + y^2 - 2py + p^2 = (y+p)^2$$

$$\Rightarrow x^2 + \cancel{y^2} - 2py + \cancel{p^2} = \cancel{y^2} + 2py + \cancel{p^2}$$

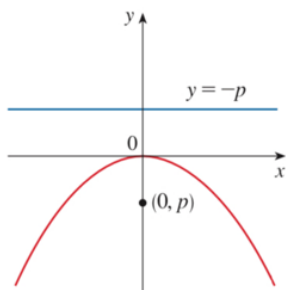
$$\Rightarrow x^2 = 4p \cdot y \quad \leftarrow \text{from here, we can say that}$$

$(0, p)$ is the focus and $y = -p$ is the directrix.

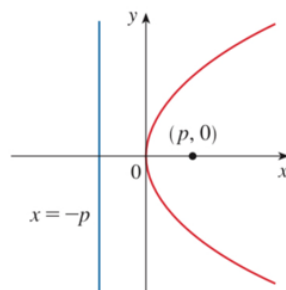
The following are some different orientations of the same parabola and their equations:



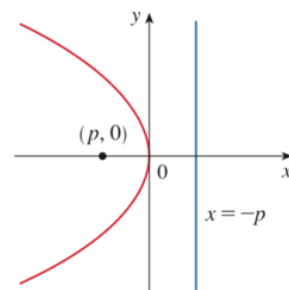
(a) $x^2 = 4py, p > 0$



(b) $x^2 = 4py, p < 0$



(c) $y^2 = 4px, p > 0$



(d) $y^2 = 4px, p < 0$

$\curvearrowright p$ is positive

$\curvearrowright p$ is negative

axis of the parabola is the x-axis

$\curvearrowright p$ is positive

$\curvearrowright p$ is negative

axis of the parabola is the x-axis

ⓔ.g. Find the focus and directrix of the parabola:

$$y^2 + 10x = 0$$

and sketch the graph.

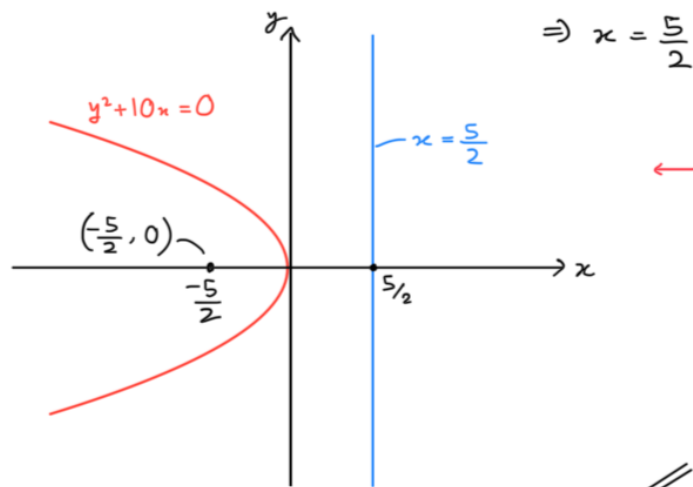
Solⁿ

$$y^2 + 10x = 0 \Rightarrow y^2 = -10x$$

$$\Rightarrow y^2 = 4\left(-\frac{5}{2}\right)x$$

So $p = -\frac{5}{2}$ and the focus is $\left(-\frac{5}{2}, 0\right)$

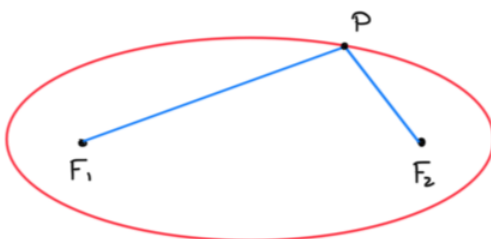
and the directrix is $x = -\left(-\frac{5}{2}\right)$



← always label these 3

things: - curve
- focus
- directrix

Ellipse: Any point P on an ellipse is such that the sum of F_1P (distance between P and the first focus) and F_2P (distance between P and the second focus) is always the same.



i.e. So for any point on the ellipse, P :

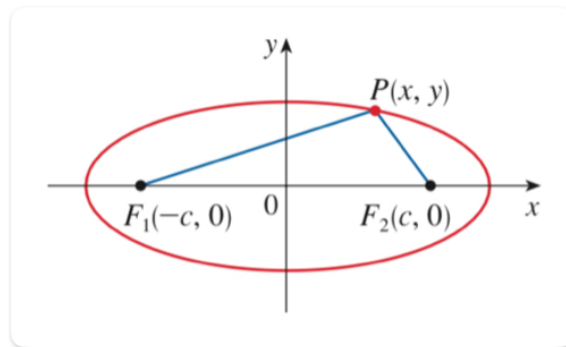
$$|F_1P| + |F_2P| = \text{constant}$$

↳ Major axis: The straight line which passes through both of the foci.

↳ Center of the ellipse: The point on the major axis that's on the midpoint between the foci.

↳ Minor axis: The straight line perpendicular to the major axis and passing through the center.

→ Consider a simple ellipse with center at the origin:



↳ To derive the equation of the ellipse, we use the property of ellipses that:

$$|F_1P| + |F_2P| = \text{constant}$$

Let the constant be $2a$.

$$\therefore |F_1P| + |F_2P| = 2a$$

$$\Rightarrow \sqrt{(x-(-c))^2 + (y-0)^2} + \sqrt{(x-c)^2 + (y-0)^2} = 2a$$

$$\Rightarrow \sqrt{(x+c)^2 + y^2} + \sqrt{(x-c)^2 + y^2} = 2a$$

$$\Rightarrow \sqrt{(x-c)^2 + y^2} = 2a - \sqrt{(x+c)^2 + y^2}$$

squaring both sides $\rightarrow$ $\Rightarrow (x-c)^2 + y^2 = 4a^2 - 4a\sqrt{(x+c)^2 + y^2} + [(x+c)^2 + y^2]$

$$\Rightarrow \cancel{x^2} - 2xc + \cancel{c^2} + \cancel{y^2} = 4a^2 - 4a\sqrt{(x+c)^2 + y^2} + \cancel{x^2} + 2xc + \cancel{c^2} + \cancel{y^2}$$

$$\Rightarrow 0 = 4a^2 - 4a\sqrt{(x+c)^2 + y^2} + 4xc$$

$$\Rightarrow a\sqrt{(x+c)^2 + y^2} = a^2 + xc \quad \leftarrow \text{dividing both sides by 4}$$

squaring both sides $\rightarrow$ $\Rightarrow a^2[(x+c)^2 + y^2] = a^4 + 2a^2xc + x^2c^2$

$$\Rightarrow a^2[x^2 + 2xc + c^2 + y^2] = a^4 + 2a^2xc + x^2c^2$$

$$\Rightarrow a^2x^2 + \cancel{2a^2cx} + a^2c^2 + a^2y^2 = a^2 \cdot a^2 + \cancel{2a^2cx} + c^2x^2$$

$$\Rightarrow (a^2 - c^2)x^2 + a^2y^2 = a^2(a^2 - c^2)$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1 \quad \leftarrow \text{dividing both sides by } a^2(a^2 - c^2)$$

Let $b^2 = a^2 - c^2$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$\leftarrow$ From here, we can see that $b^2 = a^2 - c^2 < a^2 \Rightarrow b < a$

The x-intercepts are $x = -a$ and $x = +a$ and

the y-intercepts are $y = -b$ and $y = +b$.

$\therefore$ Length of the major axis $= 2a$

Length of the minor axis $= 2b$

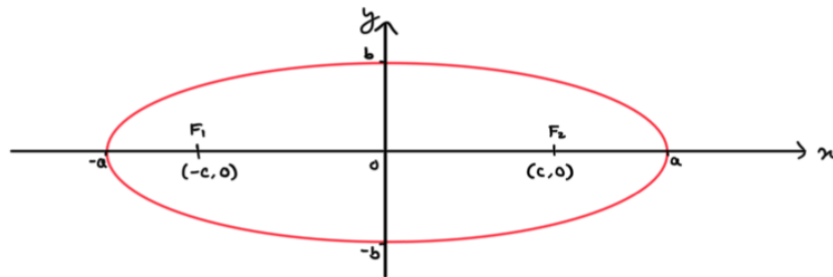
→ Foci always lie on the major axis

$$\text{and: } \begin{cases} c^2 = a^2 - b^2 & \text{when } b < a \rightarrow F_1(-c, 0), F_2(c, 0) \\ c^2 = b^2 - a^2 & \text{when } a < b \rightarrow F_1(0, -c), F_2(0, c) \end{cases}$$

→ When $b < a \rightarrow$ Major axis: x -axis $[(-a, 0) \text{ to } (a, 0)]$

Minor axis: y -axis $[(0, -b) \text{ to } (0, b)]$

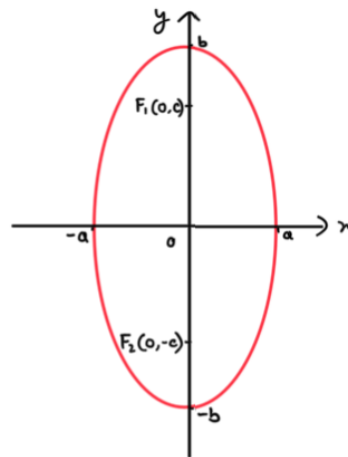
Vertices are $(-a, 0)$ and $(a, 0)$ →



When $a < b \rightarrow$ Major axis: y -axis $[(0, -b) \text{ to } (0, b)]$

Minor axis: x -axis $[(-a, 0) \text{ to } (a, 0)]$

Vertices are $(0, -b)$ and $(0, b)$ →



* If $a=b$, then this just becomes a circle.

* Vertices of an ellipse lie on the major axis.

(e.g.) Sketch the graph of $9x^2 + 16y^2 = 144$ and locate the foci.

Solⁿ Since, in the standard form of an ellipse, there's only 1 in the RHS, we divide both sides by 144:

$$\frac{9x^2}{144} + \frac{16y^2}{144} = \frac{144}{144}$$

$$\Rightarrow \frac{x^2}{16} + \frac{y^2}{9} = 1 \quad \leftarrow \text{This gives us } a^2 = 16 \text{ and } b^2 = 9$$

$$\text{As } a > b, \quad c^2 = a^2 - b^2 \quad \therefore a = 4 \text{ and } b = 3$$

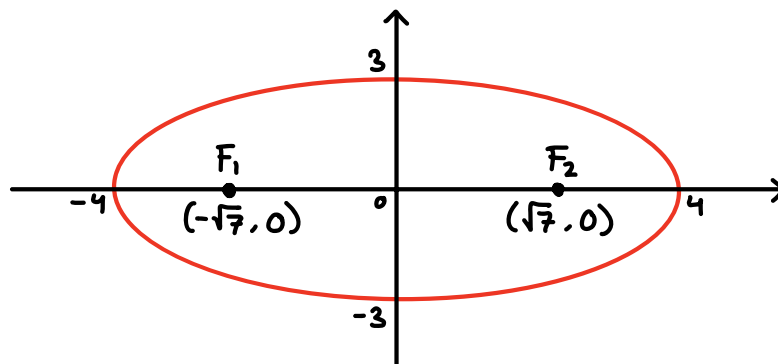
$$\Rightarrow c^2 = 16 - 9$$

$$\Rightarrow c = \sqrt{7}$$

Also, since $a > b$, the major axis is the horizontal axis.

$\therefore$ The foci lie on the horizontal axis.

So the foci are: $(-\sqrt{7}, 0)$ and $(+\sqrt{7}, 0)$.



(e.g.) Find the equation of an ellipse with foci $(0, \pm 2)$ and vertices $(0, \pm 3)$.

Solⁿ Since the foci are $(0, -2)$ and $(0, 2)$, we have $c=2$.

Also as they are on the vertical axis, we can say that it's the major axis $\longrightarrow a < b$ and $c^2 = b^2 - a^2$.

The vertices are $(0, -3)$ and $(0, 3)$, so $b=3$.

$$\therefore c^2 = b^2 - a^2$$

$$\Rightarrow 2^2 = 3^2 - a^2$$

$$\Rightarrow 4 = 9 - a^2$$

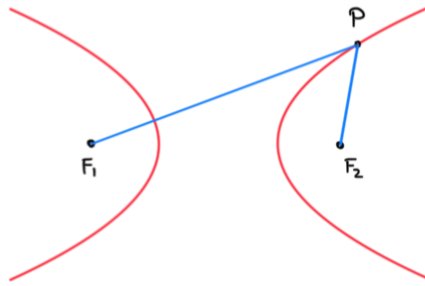
$$\Rightarrow a = \sqrt{5}$$

Hence, eqⁿ of the ellipse is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{x^2}{5} + \frac{y^2}{9} = 1 //$$

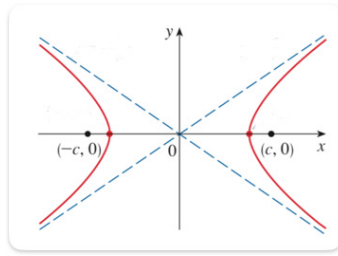
Hyperbola: Any point P on a hyperbola is such that the difference in distance to P from F_1 (the first focus) and F_2 (the second focus) is always constant.



i.e. $|F_1P| - |F_2P| = \text{constant}$

- ↳ Focal axis: Line segment joining the foci of the hyperbola.
- ↳ Center of the hyperbola: Point on the focal axis of the hyperbola that's at the midpoint between the foci.
- ↳ Conjugate axis: Line perpendicular to the focal axis and passing through the center of the hyperbola.
- ↳ Vertices: Intersections of the hyperbola with the focal axes.

→ Consider a simple hyperbola with center at the origin:



To derive the equation of the hyperbola, we use the property:

$$|F_1P| - |F_2P| = \text{constant}$$

→ Let the constant be $2a$:

$$\therefore |F_1P| - |F_2P| = 2a$$

$$\Rightarrow \sqrt{(x - (-c))^2 + (y - 0)^2} - \sqrt{(x - c)^2 + (y - 0)^2} = 2a$$

$$\Rightarrow \sqrt{(x+c)^2 + y^2} = 2a + \sqrt{(x-c)^2 + y^2}$$

$$\Rightarrow (x+c)^2 + \cancel{y^2} = 4a^2 + 4a\sqrt{(x-c)^2 + y^2} + (x-c)^2 + \cancel{y^2}$$

$$\Rightarrow \cancel{x^2} + 2cx + \cancel{c^2} = 4a^2 + 4a\sqrt{x^2 - 2cx + c^2 + y^2} + \cancel{x^2} - 2cx + \cancel{c^2}$$

$$\Rightarrow 4cx - 4a^2 = 4a\sqrt{x^2 - 2cx + c^2 + y^2}$$

$$\Rightarrow cx - a^2 = a\sqrt{x^2 - 2cx + c^2 + y^2}$$

$$\Rightarrow (cx - a^2)^2 = a^2(x^2 - 2cx + c^2 + y^2)$$

$$\Rightarrow c^2x^2 - \cancel{2a^2cx} + a^4 = a^2x^2 - \cancel{2a^2cx} + a^2c^2 + a^2y^2$$

$$\Rightarrow x^2(c^2 - a^2) + a^2(a^2 - c^2) - a^2y^2 = 0$$

$$\Rightarrow x^2(c^2 - a^2) + a^2(a^2 - c^2) - a^2y^2 = 0$$

$$\Rightarrow x^2(c^2 - a^2) - a^2y^2 = a^2(c^2 - a^2) \quad \leftarrow \text{Dividing both sides by } a^2(c^2 - a^2)$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{c^2 - a^2} = 1$$

Let $b^2 = c^2 - a^2$:

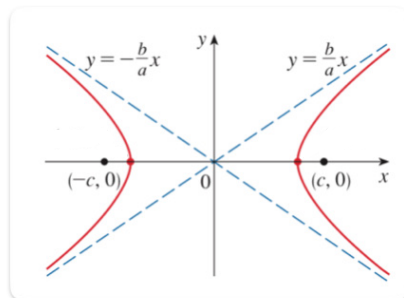
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$\leftarrow$ From here we can see that $b^2 = c^2 - a^2$
 $\Rightarrow c^2 = a^2 + b^2$

* It does not matter if $b < a$ or $b > a$.

The x-intercepts are $x = a$ and $x = -a$
 and there are no y-intercepts.

$\rightarrow$ Hyperbolas have two asymptotes as follows:



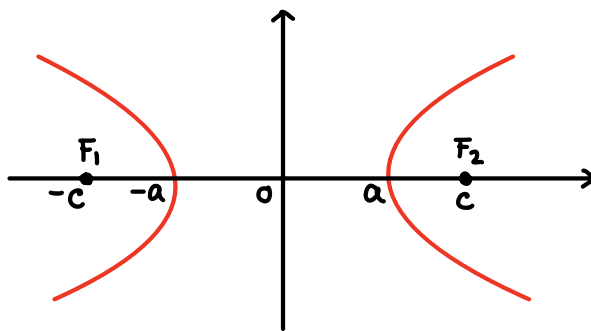
The asymptotes of hyperbolae with equations

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{and} \quad \frac{y^2}{b^2} - \frac{x^2}{a^2} = 1 \quad \text{both are given by:}$$

$$y = \pm \frac{b}{a} x$$

→ For hyperbolae with equation: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

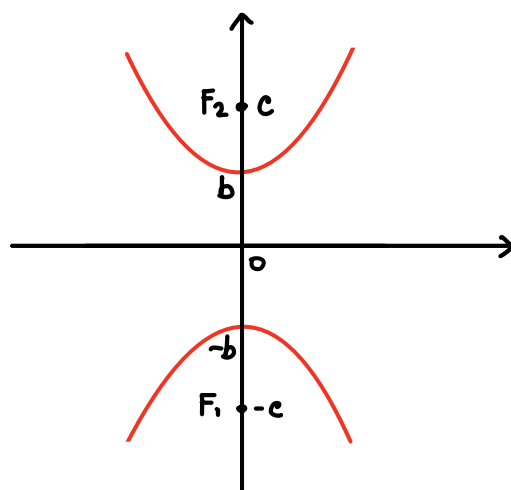
Focal axis
is horizontal



← Foci: $(-c, 0)$
and $(c, 0)$
Vertices: $(-a, 0)$
and $(a, 0)$

→ For hyperbolae with equation: $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$

Focal axis
is vertical



← Foci: $(0, -c)$
and $(0, c)$
Vertices: $(0, -b)$
and $(0, b)$

→ In both cases, $c^2 = a^2 + b^2$.

Ⓔ.g. Find the foci and the asymptotes of the hyperbola
 $9x^2 - 16y^2 = 144$ and sketch its graph.

Solⁿ

$$\Rightarrow \frac{x^2}{16} - \frac{y^2}{9} = 1 \quad \leftarrow \text{Dividing both sides by 144.}$$

From the above eqⁿ, we can

see that $a^2 = 16$ and $b^2 = 9 \quad \therefore a = 4, b = 3$.

Additionally, this form of the equation means that the focal axis is horizontal.

So the vertices are also on the horizontal axis: $(-4, 0)$ and $(4, 0)$.

$$\text{Also, } c^2 = a^2 + b^2 = 16 + 9$$

$$\Rightarrow c = \sqrt{25} = 5 \quad \leftarrow \text{Foci are at } (-5, 0) \text{ and } (5, 0).$$

$$\text{Asymptotes are at } y = \pm \frac{b}{a}x = \pm \frac{3}{4}x$$

$$\therefore y = -\frac{3}{4}x \text{ and } y = +\frac{3}{4}x.$$

$\therefore$ Sketching the graph:

